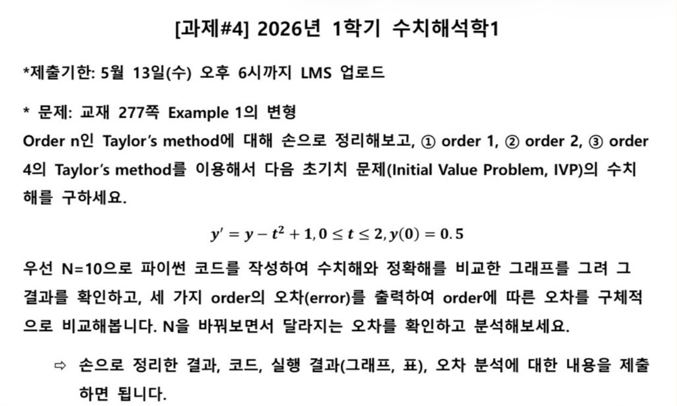


[과제#4] 2026년 1학기 수치해석학1

*제출기한: 5월 13일(수) 오후 6시까지 LMS 업로드

* 문제: 교재 277쪽 Example 1의 변형
Order n인 Taylor's method에 대해 손으로 정리해보고, ① order 1, ② order 2, ③ order 4의 Taylor's method를 이용해서 다음 초기치 문제(Initial Value Problem, IVP)의 수치해를 구하세요.

$$y' = y - t^2 + 1, 0 \le t \le 2, y(0) = 0.5$$

우선 N=10으로 파이썬 코드를 작성하여 수치해와 정확해를 비교한 그래프를 그려 그 결과를 확인하고, 세 가지 order의 오차(error)를 출력하여 order에 따른 오차를 구체적으로 비교해봅니다. N을 바꿔보면서 달라지는 오차를 확인하고 분석해보세요.

⇨ 손으로 정리한 결과, 코드, 실행 결과(그래프, 표), 오차 분석에 대한 내용을 제출하면 됩니다.

학번 : 2022114514

이름 : 김민래.

Order n, Taylor' method.

$t_{i+1} = t_i + h$.

$y(t_{i+1})$

$$= y(t_i+h) = y(t_i) + y'(t_i)h + \frac{y''(t_i)h^2}{2!} + \frac{y'''(t_i)h^3}{3!} \cdots\cdots \frac{y^{(n)}(t_i)h^n}{n!} \cdots\cdots\cdots$$

$$= y(t_i) + y'(t_i)h + \frac{y''(t_i)h^2}{2!} + \frac{y'''(t_i)h^3}{3!} \cdots \frac{y^{(n)}(t_i)}{n!}h^n + O(h^{n+1})$$

$$\approx w_n(t_i) = y(t_i) + y'(t_i)h + \frac{y''(t_i)}{2!}h^2 + \frac{y'''(t_i)}{3!}h^3 \cdots\cdots \frac{y^{(n)}(t_i)}{n!}h^n$$

$O(h^{n+1})$의 오차 발생.

$\Rightarrow$ $w(t_0) = y(t_0)$.

$y'(t_i) = f(t_i, y_i) \approx f(t_i, w(t_i)) = w_n(t_i) - t_i^2 + 1 = w_n'(t_i)$ ……………… ①

$y''(t_i) = f'(t_i, y_i) \approx f'(t_i, w(t_i)) = w_n'(t_i) - 2t_i = w_n''(t_i) = w(t_i) - t_i^2 - 2t_i + 1$ ---- ②

$y'''(t_i) = f''(t_i, y_i) \approx f''(t_i, w(t_i)) = w_n''(t_i) - 2 = w_n'''(t_i) = w(t_i) - t_i^2 - 2t_i - 1$ ----- ③

$y''''(t_i) = f'''(t_i, y_i) \approx f'''(t_i, w(t_i)) = w_n'''(t_i) = w_n''''(t_i) = w(t_i) - t_i^2 - 2t_i - 1$ --- ④

3차식부터 동일함.

order - 1 n=1.

$$y(t_i+h) = y(t_i) + y'(t_i)h + O(h^2)$$

$$= y(t_i) + f(t_i, y_i)h + O(h^2)$$

$$\approx w_1(t_i+h) = w(t_i) + \underbrace{f(t_i, w(t_i))}_{①}h.$$

오차 $O(h^2)$.

order - 2 n=2.

$$y(t_i+h) = y(t_i) + y'(t_i)h + \frac{y''(t_i)}{2!}h^2 + O(h^3)$$

오차 $O(h^3)$.

$$\approx w(t_i+h) = w(t_i) + w'(t_i)h + \frac{w''(t_i)}{2!}h^2$$

$$= w(t_i) + \Big(\underbrace{f(t_i, w(t_i))}_{①} + \frac{1}{2}\underbrace{f'(t_i, w(t_i))}_{②}h\Big)h.$$

order - 4. n=4.

$$y(t_i+h) = y(t_i) + y'(t_i)h + \frac{y''(t_i)}{2!}h^2 + \frac{y'''(t_i)}{3!}h^3 + \frac{y''''(t_i)}{4!}h^4 + O(h^5)$$

오차 : $O(h^5)$

$$\approx w(t_i+h) = w(t_i) + w'(t_i)h + \frac{w''(t_i)}{2!}h^2 + \frac{w'''(t_i)}{3!}h^3 + \frac{w''''(t_i)}{4!}h^4$$

$$= w(t_i) + \Big(\underbrace{f(t_i, w(t_i))}_{①} + \frac{1}{2}\underbrace{f'(t_i, w(t_i))}_{②}h + \frac{1}{3!}\underbrace{f''(t_i, w(t_i))}_{③}h^2 + \frac{1}{4!}\underbrace{f'''(t_i, w(t_i))}_{④}h^3\Big)h.$$

파이썬 실습 결과.

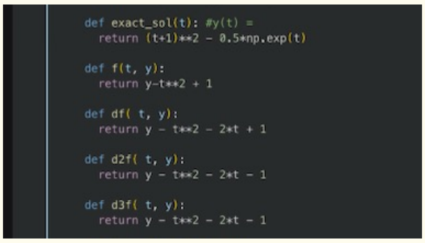
$f(t_i, w_i), f''(t_i, w_i), f'''(t_i, w_i), f''''(t_i, w_i)$ 함수정의.

$$f(t_i, w(t_i)) = w_n(t_i) - t_i^2 + 1 = \underline{w_n'(t_i)}$$ ·········· ①

$$f'(t_i, w(t_i)) = w_n'(t_i) - 2t_i = \underline{w_n''(t_i)} = w(t_i) - t_i^2 - 2t_i + 1$$ ···· ②

$$f''(t_i, w(t_i)) = w_n''(t_i) - 2 = \underline{w_n'''(t_i)} = w(t_i) - t_i^2 - 2t_i - 1$$ ····· ③

$$f'''(t_i, w(t_i)) = \underline{w_n'''(t_i)} = w_n''''(t_i) = w(t_i) - t_i^2 - 2t_i - 1$$ ···· ④



```
def exact_sol(t): #y(t) =
  return (t+1)**2 - 0.5*np.exp(t)

def f(t, y):
  return y-t**2 + 1

def df( t, y):
  return y - t**2 - 2*t + 1

def d2f( t, y):
  return y - t**2 - 2*t - 1

def d3f( t, y):
  return y - t**2 - 2*t - 1
```

Order 1, 2, 4 함수정의.

order - 1 n=1.

$$w_1(t_i+h) = w(t_i) + \underbrace{f(t_i, w(t_i))}_{\textcircled{1}}h.$$

order - 2 n=2.

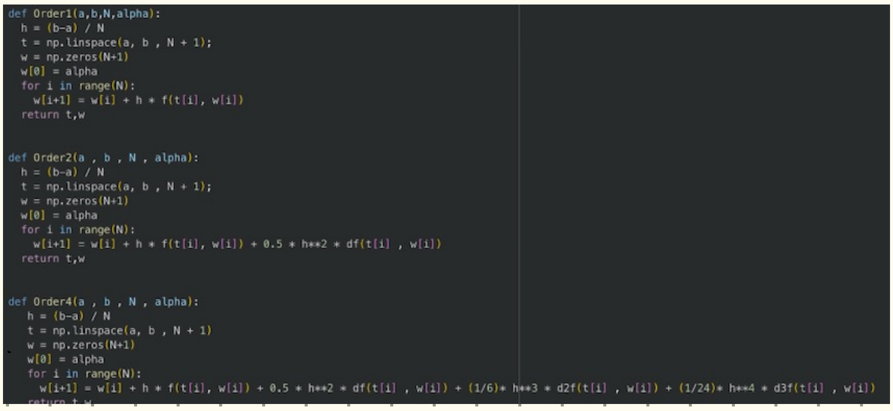
$$w(t_i+h) = w(t_i) + w'(t_i)h + \frac{w''(t_i)}{2!}h^2$$

$$= w(t_i) + \left(\underbrace{f(t_i, w(t_i))}_{\textcircled{1}} + \frac{1}{2}\underbrace{f'(t_i, w(t_i))}_{\textcircled{2}}h\right)h$$

order - 4. n=4.

$$w(t_i+h) = w(t_i) + w'(t_i)h + \frac{w''(t_i)}{2!}h^2 + \frac{w'''(t_i)}{3!}h^3 + \frac{w''''(t_i)}{4!}h^4$$

$$= w(t_i) + \left(\underbrace{f(t_i, w(t_i))}_{\textcircled{1}} + \frac{1}{2}\underbrace{f'(t_i, w(t_i))}_{\textcircled{2}}h + \frac{1}{3!}\underbrace{f''(t_i, w(t_i))}_{\textcircled{3}}h^2 + \frac{1}{4!}\underbrace{f'''(t_i, w(t_i))}_{\textcircled{4}}h^3\right)h.$$

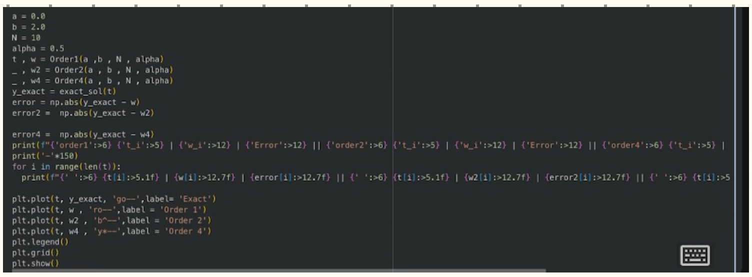


```
def Order1(a,b,N,alpha):
  h = (b-a) / N
  t = np.linspace(a, b , N + 1);
  w = np.zeros(N+1)
  w[0] = alpha
  for i in range(N):
    w[i+1] = w[i] + h * f(t[i], w[i])
  return t,w


def Order2(a , b , N , alpha):
  h = (b-a) / N
  t = np.linspace(a, b , N + 1);
  w = np.zeros(N+1)
  w[0] = alpha
  for i in range(N):
    w[i+1] = w[i] + h * f(t[i], w[i]) + 0.5 * h**2 * df(t[i] , w[i])
  return t,w


def Order4(a , b , N , alpha):
   h = (b-a) / N
   t = np.linspace(a, b , N + 1)
   w = np.zeros(N+1)
   w[0] = alpha
   for i in range(N):
     w[i+1] = w[i] + h * f(t[i], w[i]) + 0.5 * h**2 * df(t[i] , w[i]) + (1/6)* h**3 * d2f(t[i] , w[i]) + (1/24)* h**4 * d3f(t[i] , w[i])
   return t,w
```

출력코드.



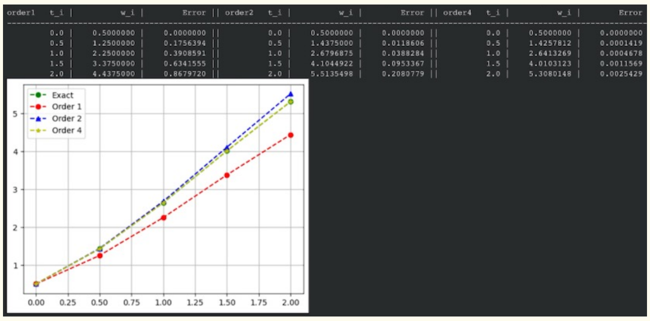
```
a = 0.0
b = 2.0
N = 10
alpha = 0.5
t , w = Order1(a ,b , N , alpha)
_ , w2 = Order2(a , b , N , alpha)
_ , w4 = Order4(a , b , N , alpha)
y_exact = exact_sol(t)
error = np.abs(y_exact - w)
error2 =  np.abs(y_exact - w2)

error4 =  np.abs(y_exact - w4)
print(f"{'order1':>6} {'t_i':>5} | {'w_i':>12} | {'Error':>12} || {'order2':>6} {'t_i':>5} | {'w_i':>12} | {'Error':>12} || {'order4':>6} {'t_i':>5} |
print('-'*150)
for i in range(len(t)):
  print(f"{' ':>6} {t[i]:>5.1f} | {w[i]:>12.7f} | {error[i]:>12.7f} || {' ':>6} {t[i]:>5.1f} | {w2[i]:>12.7f} | {error2[i]:>12.7f} || {' ':>6} {t[i]:>5

plt.plot(t, y_exact, 'go--',label= 'Exact')
plt.plot(t, w , 'ro--',label = 'Order 1')
plt.plot(t, w2 , 'b^--',label = 'Order 2')
plt.plot(t, w4 , 'y*--',label = 'Order 4')
plt.legend()
plt.grid()
plt.show()
```

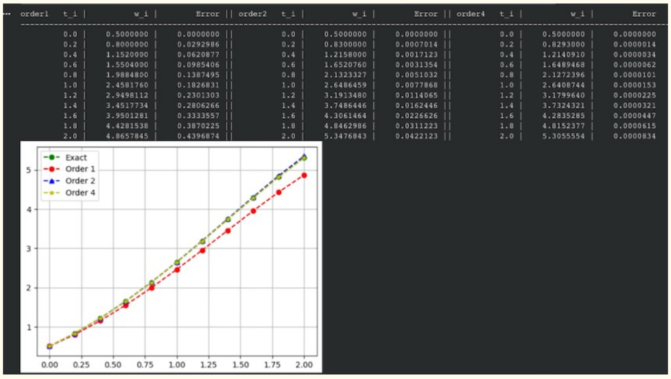
---

n=4.



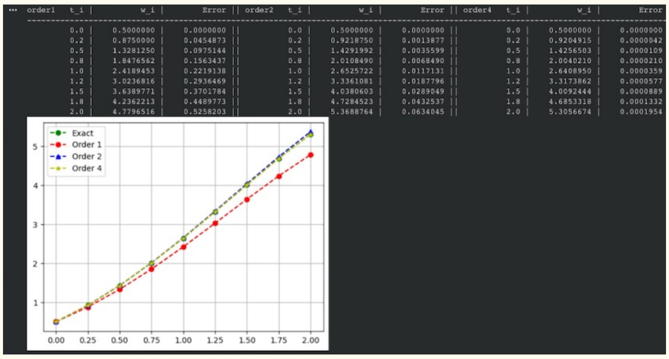
```
order1   t_i |          w_i |        Error || order2   t_i |          w_i |        Error || order4   t_i |          w_i |        Error
------------------------------------------------------------------------------------------------------------------------------------------------------
         0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000
         0.5 |    1.2500000 |    0.1756394 ||          0.5 |    1.4375000 |    0.0118606 ||          0.5 |    1.4257812 |    0.0001419
         1.0 |    2.2500000 |    0.3908591 ||          1.0 |    2.6796875 |    0.0388284 ||          1.0 |    2.6413269 |    0.0004678
         1.5 |    3.3750000 |    0.6341555 ||          1.5 |    4.1044922 |    0.0953367 ||          1.5 |    4.0103123 |    0.0011569
         2.0 |    4.4375000 |    0.8679720 ||          2.0 |    5.5135498 |    0.2080779 ||          2.0 |    5.3080148 |    0.0025429
```

n=10.



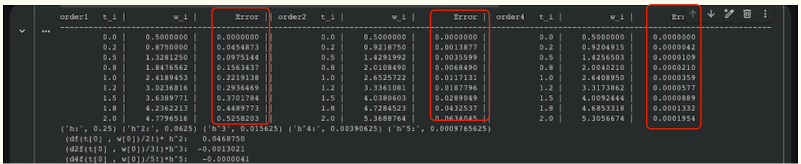
```
order1   t_i |          w_i |        Error || order2   t_i |          w_i |        Error || order4   t_i |          w_i |        Error
------------------------------------------------------------------------------------------------------------------------------------------------------
         0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000
         0.2 |    0.8000000 |    0.0292986 ||          0.2 |    0.8300000 |    0.0007014 ||          0.2 |    0.8293000 |    0.0000014
         0.4 |    1.1520000 |    0.0620877 ||          0.4 |    1.2158000 |    0.0017123 ||          0.4 |    1.2140910 |    0.0000034
         0.6 |    1.5504000 |    0.0985406 ||          0.6 |    1.6520760 |    0.0031354 ||          0.6 |    1.6489468 |    0.0000062
         0.8 |    1.9884800 |    0.1387495 ||          0.8 |    2.1323327 |    0.0051032 ||          0.8 |    2.1272396 |    0.0000101
         1.0 |    2.4581760 |    0.1826831 ||          1.0 |    2.6486459 |    0.0077868 ||          1.0 |    2.6408744 |    0.0000153
         1.2 |    2.9498112 |    0.2301303 ||          1.2 |    3.1913480 |    0.0114065 ||          1.2 |    3.1799640 |    0.0000225
         1.4 |    3.4517734 |    0.2806266 ||          1.4 |    3.7486446 |    0.0162446 ||          1.4 |    3.7324321 |    0.0000321
         1.6 |    3.9501281 |    0.3333557 ||          1.6 |    4.3061464 |    0.0226626 ||          1.6 |    4.2835285 |    0.0000447
         1.8 |    4.4281538 |    0.3870225 ||          1.8 |    4.8462986 |    0.0311223 ||          1.8 |    4.8152377 |    0.0000615
         2.0 |    4.8657845 |    0.4396874 ||          2.0 |    5.3476843 |    0.0422123 ||          2.0 |    5.3055554 |    0.0000834
```

n=8.



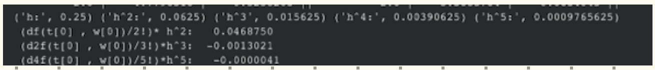
```
order1   t_i |          w_i |        Error || order2   t_i |          w_i |        Error || order4   t_i |          w_i |        Error
------------------------------------------------------------------------------------------------------------------------------------------------------
         0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000
         0.2 |    0.8750000 |    0.0454873 ||          0.2 |    0.9218750 |    0.0013877 ||          0.2 |    0.9204915 |    0.0000042
         0.5 |    1.3281250 |    0.0975144 ||          0.5 |    1.4291992 |    0.0035599 ||          0.5 |    1.4256503 |    0.0000109
         0.8 |    1.8476562 |    0.1563437 ||          0.8 |    2.0108490 |    0.0068490 ||          0.8 |    2.0040210 |    0.0000210
         1.0 |    2.4189453 |    0.2219138 ||          1.0 |    2.6525722 |    0.0117131 ||          1.0 |    2.6408950 |    0.0000359
         1.2 |    3.0236816 |    0.2936469 ||          1.2 |    3.3361081 |    0.0187796 ||          1.2 |    3.3173862 |    0.0000577
         1.5 |    3.6389771 |    0.3701784 ||          1.5 |    4.0380603 |    0.0289049 ||          1.5 |    4.0092444 |    0.0000889
         1.8 |    4.2362213 |    0.4489773 ||          1.8 |    4.7284523 |    0.0432537 ||          1.8 |    4.6853318 |    0.0001332
         2.0 |    4.7796516 |    0.5258203 ||          2.0 |    5.3688764 |    0.0634045 ||          2.0 |    5.3056674 |    0.0001954
```

오차 분석.



```
order1   t_i |          w_i |        Error || order2   t_i |          w_i |        Error || order4   t_i |          w_i |        Er
-------------------------------------------------------------------------------------------------------------------------------------
         0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000
         0.2 |    0.8750000 |    0.0454873 ||          0.2 |    0.9218750 |    0.0013877 ||          0.2 |    0.9204915 |    0.0000042
         0.5 |    1.3281250 |    0.0975144 ||          0.5 |    1.4291992 |    0.0035599 ||          0.5 |    1.4256503 |    0.0000109
         0.8 |    1.8476562 |    0.1563437 ||          0.8 |    2.0108490 |    0.0068490 ||          0.8 |    2.0040210 |    0.0000210
         1.0 |    2.4189453 |    0.2219138 ||          1.0 |    2.6525722 |    0.0117131 ||          1.0 |    2.6408950 |    0.0000359
         1.2 |    3.0236816 |    0.2936469 ||          1.2 |    3.3361081 |    0.0187796 ||          1.2 |    3.3173862 |    0.0000577
         1.5 |    3.6389771 |    0.3701784 ||          1.5 |    4.0380603 |    0.0289049 ||          1.5 |    4.0092444 |    0.0000889
         1.8 |    4.2362213 |    0.4489773 ||          1.8 |    4.7284523 |    0.0432537 ||          1.8 |    4.6853318 |    0.0001332
         2.0 |    4.7796516 |    0.5258203 ||          2.0 |    5.3688764 |    0.0634045 ||          2.0 |    5.3056674 |    0.0001954
('h:', 0.25) ('h^2:', 0.0625) ('h^3', 0.015625) ('h^4:', 0.00390625) ('h^5:', 0.0009765625)
 (df(t[0] , w[0])/2!)* h^2:   0.0468750
 (d2f(t[0] , w[0])/3!)*h^3:  -0.0013021
 (d4f(t[0] , w[0])/5!)*h^5:   -0.0000041
```

다음과 같이 위 결과에서는 오차가 발생하며.



```
('h:', 0.25) ('h^2:', 0.0625) ('h^3', 0.015625) ('h^4:', 0.00390625) ('h^5:', 0.0009765625)
 (df(t[0] , w[0])/2!)* h^2:   0.0468750
 (d2f(t[0] , w[0])/3!)*h^3:  -0.0013021
 (d4f(t[0] , w[0])/5!)*h^5:   -0.0000041
```

그 결과는 누적되기되는데 이때 누적되는 각 스텝의 오차는.

order-1 n=1.

$$y(t_i+h) = y(t_i) + f(t_i, y_i)h + O(h^2)$$
$$\approx w_1(t_i+h) = w(t_i) + \underbrace{f(t_i, w(t_i))h}_{①}$$
오차 $O(h^2) \Rightarrow \left|\frac{y''(t_i)}{2!}h^2 \cdots\right| \approx \left|\frac{f'(t_i,y_i)}{2!}h^2\right| = 0.046875$

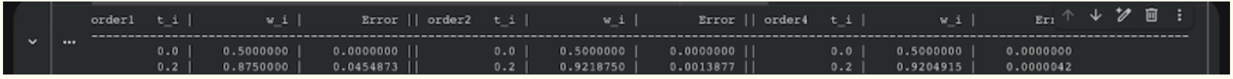
order-2 n=2.

$$y(t_i+h) = y(t_i) + y'(t_i)h + \frac{y''(t_i)}{2!}h^2 + O(h^3)$$
$$\approx w(t_i+h) = w(t_i) + w'(t_i)h + \frac{w''(t_i)}{2!}h^2$$
오차 $O(h^3) \Rightarrow \left|\frac{y'''(t_i)}{3!}h^3 \cdots\right| \approx \left|\frac{f''(t_i,y_i)}{3!}h^3\right| = 0.0013021$

order-4. n=4.

$$y(t_i+h) = y(t_i) + y'(t_i)h + \frac{y''(t_i)}{2!}h^2 + \frac{y'''(t_i)}{3!}h^3 + \frac{y''''(t_i)}{4!}h^4 + O(h^5)$$
$$\approx w(t_i+h) = w(t_i) + w'(t_i)h + \frac{w''(t_i)}{2!}h^2 + \frac{w'''(t_i)}{3!}h^3 + \frac{w''''(t_i)}{4!}h^4$$
오차 : $O(h^5) \Rightarrow \left|\frac{y^{(5)}(t_i)h^5}{5!}\cdots\right| \approx \left|\frac{f^{(4)}(t_i,y_i)}{5!}h^5\right| = 0.0000041$

order1, 2, 4 에서 위와 같이 예상된다.



```
order1   t_i |          w_i |        Error || order2   t_i |          w_i |        Error || order4   t_i |          w_i |        Er
-------------------------------------------------------------------------------------------------------------------------------------
         0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000 ||          0.0 |    0.5000000 |    0.0000000
         0.2 |    0.8750000 |    0.0454873 ||          0.2 |    0.9218750 |    0.0013877 ||          0.2 |    0.9204915 |    0.0000042
```

step1 에서 누적된 오차 와 사전에 예상된 오차는.

	step1 오차	예상. 오차.
order1	0.0454873	0.046875
order 2	0.0013877	0.013021
order 4.	0.0000042	0.0000041

다음과 같이 나타내어 지며 각 스텝의 오차는 테일러 급수 과정에서 발생한 절단 오차임을 알 수 있습니다.